PETRA ANDHIKA NATANAEL

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PROFESSIONAL SUMMARY

Informatics graduate specializing in Artificial Intelligence, with experience building end-to-end AI solutions, from data labeling, model training, and evaluation to deployment, across Computer Vision (YOLO, PyTorch, OpenCV), Edge AI (Raspberry Pi 5), and LLM/RAG. A fast learner who documents work on GitHub and is highly motivated to turn AI models from experiments into products that real users rely on.

EDUCATION

Universitas Sanata Dharma - Yogyakarta, Indonesia Sep 2022 - Sep 2026
Bachelor's Degree in Informatics, GPA 3.57/4.00

WORK EXPERIENCE

AI Engineer Intern | Beehive Drone (PT Aerotek Global Inovasi) Feb 2026 - May 2026
Project: Oil Palm Flower Ripeness Detection for Drones

- Labeled an image dataset (bounding boxes) in Roboflow for 4 oil palm flower ripeness classes.
- Trained and compared YOLOv8, YOLOv11, and YOLO26 models (Ultralytics, PyTorch); the best model reached 92.6% accuracy.
- Deployed the model to a Raspberry Pi 5 with a Sony IMX500 AI camera (Edge AI) for real-time detection on drones.

Project: Barcode, QR Code & Expiry Date Reader for Warehouses

- Built a real-time multi-threaded pipeline (Python, OpenCV): YOLO nano barcode/QR detection, tracking, zxing-cpp decoding.
- Built an expiry date reader: custom YOLO date detection, cropping, PaddleOCR PP-OCRv6 (ONNX) recognition, date parsing.
- Stored the code + expiry date per carton in CSV and a UI panel, as a proof of concept for a drone that scans warehouse racks.

ORGANIZATIONAL EXPERIENCE

Communication & Information Staff | Student Executive Board (BEM USD) Dec 2023 - Dec 2024

- Managed the planning, production, and publication of visual content (photo/video) using Figma, Canva, and CapCut for the organization's digital communication.
- Managed and redesigned the organization's website using WordPress, improving usability and keeping information up to date.

ACADEMIC PROJECT

Shuttlecock Tracking in Badminton Videos Using TrackNetV3 Aug 2026
Undergraduate Thesis, Python, PyTorch, OpenCV

- Implemented TrackNetV3 in PyTorch (Tracking Module U-Net Conv2D + Rectification Module InpaintNet Conv1D) and analyzed sequence length variations (4, 6, 8, 10 frames) on 201 amateur and professional badminton rally videos.
- Built the data pipeline (frame extraction, background estimation, Gaussian heatmap ground truth) and an end-to-end evaluation pipeline, including fixing a loss function bug and data leakage in the dataset split.
- Ran 4 progressive experiment scenarios (baseline, +background estimation, +MixUp, +Rectification Module); the full 8-frame pipeline achieved 91.75% accuracy, 94.11% recall, and 95.11% F1-score.

PERSONAL PROJECT

Petra Assistant, Personal RAG Chatbot Sep 2026
Python, Hugging Face, Ollama, OpenAI API, Gemini API, Flask, Docker

- Built an end-to-end Retrieval-Augmented Generation (RAG) pipeline with hybrid retrieval (Qwen3-Embedding-0.6B dense embeddings + BM25) and cross-encoder reranking (BAAI/bge-reranker-v2-m3) to improve retrieval precision.
- Designed a multi-provider LLM architecture (Ollama, OpenAI, Gemini) that can be swapped without changing the retrieval logic, with live document ingestion and index caching, delivered as a full-stack web app (Flask, REST API, chat UI).
- Deployed with Docker on Hugging Face Spaces and embedded in my React portfolio website as a live interactive demo; code on GitHub.

SKILLS

- **Programming Languages:** Python, SQL, Java
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Ultralytics YOLO, OpenCV, NumPy, Pandas
- **Computer Vision:** Object Detection, Object Tracking, Image Classification, OCR (PaddleOCR, EasyOCR, Tesseract), Barcode/QR Decoding
- **LLM & NLP:** RAG, Hybrid Search (BM25 + Dense), Reranking, Hugging Face, Ollama, OpenAI API, Gemini API, Text Preprocessing
- **Deployment & Tools:** Git, GitHub, Docker, Flask, REST API, Hugging Face Spaces, Vercel, Roboflow, Google Colab, Linux, ONNX, Raspberry Pi 5, Sony IMX500
- **Soft Skills:** Problem Solving, Technical Documentation, Teamwork, Fast Learner, Time Management

ACHIEVEMENTS

Selected for University Funding, Student Creativity Program (PKM) May 2025 - Jul 2025

- Designed the funded proposal “Smart Parking Management System Based on Fuzzy Logic and Optical Character Recognition for Vehicles”, combining Fuzzy Logic for decision making and OCR for license plate recognition.